

MEDOPRAZOLE®

(Omeprazole)

Composition: MEDOPRAZOLE® capsules contain 20mg Omeprazole in the form of enteric-coated pellets

Description: MEDOPRAZOLE® are white-white size '2' hard gelatin capsules

Pharmacodynamic properties: Omeprazole is a proton pump inhibitor used therapeutically to decrease gastric acid secretion. ATC code: A02BC01.

Omeprazole is activated at an acidic pH to a sulphonamide derivative that irreversibly binds to H / K ATPase, an enzyme system on the secretory surface of parietal cells. This results in inhibition of the final transport of hydrogen ions, via exchange with potassium ions, into the gastric lumen.

Omeprazole inhibits both basal and stimulated acid secretion.

Local information on the prevalence of resistance and local therapeutic guidelines should be taken into account in the choice of an appropriate combination regimen for *H. pylori* eradication therapy. Furthermore, in patients with persistent infection, potential development of secondary resistance (in patients with primary susceptible strains) to an antibacterial agent should be taken into account in the considerations for a new retreatment regimen.

During treatment with antisecretory medicinal products, serum gastrin increases in response to decreased acid secretion. Also CgA increase due to decreased gastric acidity. The increased CgA level may interfere with investigations for neuroendocrine tumours.

Available published evidence suggests that proton pump inhibitors should be discontinued between 5 days and 2 weeks prior to CgA measurements. This is to allow CgA levels that might be spuriously elevated following PPI treatment to return to reference range.

Pharmacokinetic properties: Omeprazole is rapidly absorbed when administered orally. Absorption of omeprazole is not affected by food. The absorption of omeprazole seems to be dose dependent. It has been reported that when the dosage was increased above 40mg the plasma concentrations raised in a non-linear way due to saturable first pass hepatic metabolism. When omeprazole is administered for a long period of time absorption becomes higher.

The bioavailability of omeprazole may be increased in the elderly patients and in patients with hepatic impairment. The bioavailability of omeprazole is not affected in patients with reduced renal function.

Omeprazole is completely metabolised in the liver, mainly to the hydroxy-omeprazole metabolite and to a lesser extent to the omeprazole sulfone metabolite. These metabolites are mostly (approx. 80%) excreted in the urine. The elimination half-life varies from 0.5 to 3 hours. Omeprazole is highly bound (approx.95%) to plasma proteins.

INDICATIONS

MEDOPRAZOLE® is indicated for:

- Therapy and symptomatic relief of oesophageal reflux disease including reflux oesophagitis and acid reflux disease.
- Therapy of benign gastric ulcers and duodenal ulcers.
- Therapy of acid related dyspepsia.
- Therapy and prophylaxis of benign gastric ulcers, duodenal ulcers and gastroduodenal erosions associated with non-steroidal anti-inflammatory drug (NSAID) therapy in patients where such symptoms arise during NSAID therapy or who have a previous history of such lesions, and require continuing NSAID therapy.
- Prophylaxis of acid aspiration during surgical procedures under general anaesthesia.
- Therapy of Zollinger – Ellison syndrome.
- In combination with appropriate antibacterial therapeutic regimens for the eradication of *Helicobacter pylori* and prevention of relapse of peptic ulcers in patients with *H. pylori* associated ulcers.

RECOMMENDED DOSAGE, DOSAGE SCHEDULE AND ROUTE OF ADMINISTRATION

Capsules are for oral administration only and should be swallowed whole with water. The capsules should not be crushed or chewed.

Adults:

Therapy and symptomatic relief of oesophageal reflux disease including reflux oesophagitis and acid reflux disease:

The usual recommended dosage is omeprazole 20mg once daily. In most cases clinical cure is affected after four

weeks therapy. In those patients not fully healed following an initial four weeks therapy, prolongation for a further four to eight weeks of therapy usually results in cure. In patients with reflux oesophagitis that is refractory to other therapy a dose of omeprazole 40mg once a day has been effective, with healing usually within eight weeks. If required patients may be maintained at a dosage of 20 mg once a day.

For long term management of acid reflux disease the recommended dosage is omeprazole 10 mg once a day, increasing to 20 mg once a day in the event of symptoms returning.

Therapy of benign gastric ulcers and duodenal ulcers:

The usual recommended dosage is omeprazole 20mg once a day. In most cases, patients with gastric ulcer are healed following eight weeks therapy and those with duodenal ulcers after four weeks therapy. In patients with severe or recurring disease the dosage may be increased to omeprazole 40mg once a day.

Omeprazole 20mg once a day is recommended as long term maintenance therapy in patients with a history of recurrent duodenal ulcer.

Omeprazole 10mg once a day, increasing to 20mg once a day if symptoms return, is recommended for the prevention of relapse in duodenal ulcer patients.

Recurring ulcer relapse is a greater risk in those patients with *Helicobacter pylori* infection; those aged less than 60 years; those with persistent symptoms for greater than one year, and those who smoke. Such patients should be initiated on long term therapy of omeprazole 20mg once a day, reducing to 10mg once a day if feasible.

Therapy of acid related dyspepsia:

Dosage is dependent upon the severity and persistence of the symptoms, but the usual dosage is omeprazole 10mg or 20mg once a day for two to four weeks. In cases where patients fail to respond following four weeks of therapy, or who relapse shortly after completion of therapy, further clinical investigation is mandatory.

Therapy and prophylaxis of benign gastric ulcers, duodenal ulcers and gastroduodenal erosions associated with non-steroidal anti-inflammatory drug (NSAID) therapy in patients where such symptoms arise during NSAID therapy, or who have a previous history of such lesions, and require continuing NSAID therapy:

For therapy the recommended dosage is omeprazole 20mg once a day, which usually produces rapid symptomatic relief and, for the majority of patients, cure within four weeks. If patients are not fully healed following four weeks of therapy, prolongation of therapy for an additional four weeks usually results in healing. For prophylaxis the recommended dosage is omeprazole 20mg once a day.

Prophylaxis of acid aspiration during surgical procedures under general anaesthesia:

Omeprazole 40mg the evening prior to surgery followed by a further 40mg two to six hours prior to surgery.

Therapy of Zollinger – Ellison syndrome:

Dosage requires individual titration dependent upon the clinical response and therapy should be prolonged for as long as clinically indicated. Initial dosage is omeprazole 60mg once a day, and effective control is usually achieved within the dosage range of 20mg–120mg daily. Where the daily dose exceeds 80mg it should be administered in two equally divided doses.

In combination with appropriate antibacterial therapeutic regimens for the eradication of Helicobacter pylori and prevention of relapse of peptic ulcers in patients with H. pylori associated ulcers:

Omeprazole is only effective in this condition in association with antibiotic therapy.

Omeprazole is administered as 40mg once a day or 20mg twice a day in association with one of the recommended antibiotic regimen. Patients with gastroduodenal ulcers due to *H. pylori* infection should be treated with eradication therapy with appropriate combinations of antibiotics with adequate dosing regimens. Selection of the appropriate regimen should be based on patient tolerability and therapeutic guidelines / availability of the antibiotics.

Triple therapy regimens:

MEDOPRAZOLE® with amoxicillin 500mg and metronidazole 400mg both administered three times daily for seven days.

MEDOPRAZOLE® with clarithromycin 250mg and metronidazole 400mg both administered two times daily for seven days.

MEDOPRAZOLE® with clarithromycin 250mg and tinidazole 500mg both administered two times daily for seven days.

MEDOPRAZOLE® with clarithromycin 500mg and amoxicillin 1g both administered two times daily for seven days.

Dual therapy regimens:

MEDOPRAZOLE with amoxicillin 750mg – 1g administered two times daily for fourteen days.

MEDOPRAZOLE with clarithromycin 500mg administered three times daily for fourteen days.

Should symptoms return and the patient test *H. pylori* positive the regimen may be repeated or an alternative regimen used.

For further information on the antibiotic components of the eradication therapy see the individual product data sheet.

Children:

There is no clinical experience with omeprazole in children and usage is not recommended.

Renal impairment:

No dosage adjustment is required in renal impairment.

Hepatic impairment:

The bioavailability and half-life of omeprazole is increased in patients with hepatic impairment, dosage adjustment is required and the maximum daily dose should not exceed 20mg.

CONTRAINDICATIONS

- Hypersensitivity to omeprazole, or to any of the formulation excipients (see List of Excipients).
- Pregnancy or breast feeding (see “Pregnancy and Lactation”).

SIDE EFFECTS/ ADVERSE REACTIONS

In general omeprazole is well tolerated with undesirable effects being mild and reversible upon therapy cessation. Usually they have not been severe enough to warrant therapy cessation.

Central Nervous System: Headache has been reported, in some cases severe enough to necessitate therapy cessation. Symptoms usually resolve following therapy cessation. Paraesthesia, somnolence, insomnia and vertigo have been reported. Dizziness, faintness and feeling light headed have been associated with therapy, usually resolving on termination of therapy.

Agitation, depression, hallucinations and reversible mental confusion have occurred; these have usually been associated with severely ill patients.

Dermatological: There have been reports of skin rash, pruritus and urticaria, all of which usually resolved on therapy cessation. There have been isolated reports of alopecia, angioedema, bullous eruptions, erythema multiforme and photosensitivity reactions.

Gastrointestinal Disorders: Diarrhoea has been reported, on occasion severe enough to warrant therapy cessation. Other reactions reported are nausea and / or vomiting, constipation, flatulence and abdominal distress or pain. Isolated cases of dry mouth, stomatitis and candidiasis have been reported.

Microscopic colitis: Frequency “not known”

Haematological: Isolated reports of agranulocytosis, leucopenia, pancytopenia and thrombocytopenia.

Hepatic: Increase in liver enzymes (alkaline phosphatase, ALT, AST) have been reported. Isolated reports of hepatitis, with or without jaundice, rare reports of hepatic failure.

Immunological: Isolated reports of anaphylactic shock, bronchospasm, fever and malaise.

Metabolic: Isolated reports of excessive sweating, peripheral oedema and, in patients with pre-existing severe hepatic disease, encephalopathy. Isolated reports of gynaecomastia.

Musculoskeletal: Reports of arthritic and myalgic symptoms exist that have usually resolved upon therapy cessation.

Renal: Rare reports of interstitial nephritis.

Sensory: Isolated reports of blurred vision and taste disturbances.

Subacute Cutaneous Lupus Erythematosus (SCLE): Skin and subcutaneous tissue disorders. Frequency “not known”: Subacute cutaneous lupus erythematosus

Interstitial Nephritis: Renal and urinary disorders: Interstitial nephritis

Hypomagnesaemia: Metabolism and nutritional disorders. Frequency “not known”: hypomagnesaemia

Fracture: Musculoskeletal disorders. Frequency “uncommon”: Fracture of the hip, wrist or spine.

Clostridium Difficile Diarrhea: Infections & infestations: Clostridium difficile associated diarrhea.

Fundic Gland Polyps (Benign): Gastrointestinal disorders. Frequency “common”: Fundic gland polyps (benign)

Vitamin B12 Deficiency: Metabolic/Nutritional: Vitamin B12 deficiency

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

Omeprazole has no known effects on the ability to drive or operate machinery. However it may cause dizziness, tiredness, vertigo, mental confusion and blurred vision (see “Adverse Reactions”). Patients should be cautioned not to drive or operate machinery until they are sure they are not affected.

WARNINGS & PRECAUTIONS

Omeprazole is metabolised extensively by the liver and dosage reduction is necessary in patients with hepatic impairment. Patients with severe hepatic disease may be more likely to suffer encephalopathy (see “Side Effects / Adverse Reactions”), and caution and close observation should be maintained in such patients.

In the therapy of gastric ulcer or acid related dyspepsia the possibility of gastric malignancy must be excluded prior to initiation of therapy, as therapy may mask symptoms and delay diagnosis.

Before using MEDOPRAZOLE in combination therapy with antibiotics for *H. pylori* eradication, the prescriber should refer to the full prescribing information of the respective antibiotics for guidance.

Interference with laboratory tests

Increased Chromogranin A (CgA) level may interfere with investigations for neuroendocrine tumours. If the patient(s) are due to have a test on Chromogranin A level, Medoprazole treatment should be stopped for at least 5 days before CgA measurements to avoid this interference (see section Pharmacodynamic). If CgA and gastrin levels have not returned to reference range after initial measurement, measurements should be repeated 14 days after cessation of proton inhibitor treatment.

Regular Surveillance

Patients on proton pump inhibitor treatment (particularly those treated for long term) should be kept under regular surveillance.

Subacute Cutaneous Lupus Erythematosus (SCLE)

Proton pump inhibitors are associated with very infrequent cases of subacute cutaneous lupus erythematosus (SCLE). If lesions occur, especially in sun-exposed areas of the skin, and if accompanied by arthralgia, the patient should seek medical help promptly and the health care professional should consider stopping Medoprazole. SCLE after previous treatment with a proton pump inhibitor may increase the risk of SCLE with other proton pump inhibitors.

Hypomagnesaemia

Severe hypomagnesaemia has been reported in patients treated with PPI like Medoprazole for at least three months, and in most cases for a year. Serious manifestations of hypomagnesaemia such as fatigue, tetany, delirium, convulsions, dizziness and ventricular arrhythmia can occur but they may begin insidiously and be overlooked. In most affected patients, hypomagnesaemia improved after magnesium replacement and discontinuation of the PPI.

For patients expected to be on prolonged treatment or who take PPI with digoxin or drugs that may cause hypomagnesaemia (e.g., diuretics), health care professionals should consider measuring magnesium levels before starting PPI treatment and periodically during treatment.

Fracture

Proton pump inhibitors, especially if used in high doses and over long durations (>1 year), may modestly increase

the risk of hip, wrist and spine fracture, predominantly in the elderly or in presence of other recognized risk factors. Observational studies suggest that proton pump inhibitors may increase the overall risk of fracture by 10-40%. Some of this increase may be due to other risk factors. Patients at risk of osteoporosis should receive care according to current clinical guidelines and they should have an adequate intake of vitamin D and calcium.

Clostridium Difficile Diarrhea

Published observational studies suggest that PPI therapy may be associated with an increased risk of Clostridium difficile associated diarrhea, especially in hospitalized patients. This diagnosis should be considered for diarrhea that does not improve. Patients should use the lowest dose and shortest duration of PPI therapy appropriate to the condition being treated.

Vitamin B12 Deficiency

Daily treatment with any acid-suppressing medications over a long period of time (e.g., longer than 3 years) may lead to malabsorption of cyanocobalamin (vitamin B12) caused by hypo- or achlorhydria. Rare reports of cyanocobalamin deficiency occurring with acid-suppressing therapy have been reported in the literature. This diagnosis should be considered if clinical symptoms consistent with cyanocobalamin deficiency are observed.

INTERACTIONS WITH OTHER MEDICAMENTS AND OTHER FORMS OF INTERACTION

Drugs:

General: As a result of omeprazole causing elevation of gastric pH there is the potential for affecting the bioavailability of any drug where absorption is pH dependent. Omeprazole may also prevent, or reduce, the degradation of acid labile drugs.

Omeprazole is metabolised in the liver by the cytochrome P450 system and there is the potential for interactions with other drugs also metabolised by this route.

Clarithromycin: Concomitant administration with omeprazole results in enhanced plasma concentrations of both drugs. This is regarded as a clinically helpful interaction during combination therapy for *H. pylori* eradication.

Digoxin: The bioavailability of digoxin increased by about 10% during simultaneous treatment with omeprazole in healthy volunteers as a result of elevated gastric pH.

Ketoconazole: Due to elevated gastric pH the absorption of ketoconazole may be reduced during omeprazole therapy, in common with other inhibitors of acid secretion.

Phenytoin: Due to omeprazole metabolism via the cytochrome P450 system it may delay the elimination of phenytoin leading to elevated phenytoin levels. Monitoring of patients concomitantly receiving phenytoin is recommended and phenytoin dosage reduction may be required.

Food: The absorption of omeprazole is unaffected by food or alcohol.

PREGNANCY AND LACTATION

Studies in rat did not demonstrate any toxic effect or inhibition of reproductive performance, and did not show teratogenic potential. Dose related embryo and foetal toxicity and post natal development toxicity were observed in rat offspring of dosed parents. In the rabbit omeprazole produced dose related increases in embryo lethality, foetal resorption and pregnancy disruption (see "Preclinical Safety Data"). Such studies are not always predicative of the effect in man.

There are no adequate and well controlled studies in man. The use of omeprazole in pregnancy is contraindicated (see also "Contraindications"). No data is available on the excretion of omeprazole in breast milk. Omeprazole has been shown to cause tumorigenic and carcinogenic effects in animals. It is recommended that breast-feeding be discontinued during omeprazole therapy (see also "Contraindications").

SYMPTOMS AND TREATMENT FOR OVERDOSAGE AND ANTIDOTE

There is no clinical experience with deliberate overdose available. In animal, toxic doses of omeprazole caused sedation, convulsions, decreased activity, depressed body temperature, depressed respiration rate and increased depth of respiration. There is no specific antidote for omeprazole and it is not readily dialysable. Treatment of overdose should be symptomatic and supportive.

PACKING/ PACK SIZES

The capsules are packaged in security plastic (PVC) containers, sealed with a PVC cap bearing a label with the name of the product, MEDOPRAZOLE®, the strength 20mg. Packs of 20 capsules are available.

STORAGE CONDITIONS

Store below 30°C in the original package, protected from light

SHELF LIFE: 3 years

Instruction for Use/Handling: A sachet containing desiccant is included into the bottles. Do not swallow it.

LIST OF EXCIPIENTS

Diethylphthalate, Sodium lauryl sulphate, Sodium dibasic phosphate, Hydroxypropylmethyl cellulose phthalate, Hydropropyl cellulose, Lactose, Hydroxypropylmethyl cellulose, Maize starch, Glucose, Sucrose

MANUFACTURER

MEDOCHEMIE LTD (Central Factory), 1 – 10, Constantinoupoleos Street, 3011 Limassol, Cyprus

PRODUCT REGISTRATION HOLDER

KOMEDIC SDN BHD, 4, Jalan PJS 11/14, Bandar Sunway, 46150 Petaling Jaya, Selangor, Malaysia.

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